Attribution from BibTeX

This statement imports its bibliography from a BibTeX file. Mathematical definitions come from the separately imported linear algebra module.

Result 1 (similarity preserves eigenvectors). [stewart2001, Theorem 1.10, page 9]
Let n be a natural number. Let A and S be $n \times n$ matrices over $\mathbb{C}$. Let x be a vector of length n over $\mathbb{C}$. Let μ be a complex number. Assume $\det(S) \neq 0$ (`invertibility`). Assume $x \neq 0$ (`nonzero`). Assume $A\,x = \mu\,x$ (`eigenpair`). Let $y = S^{-1}\,x$. Then

$$y \neq 0 \text{ and } S^{-1}\,A\,S\,y = \mu\,y.$$

Adapted from the cited source. Scope: eigenvector part.

Imported source

stewart2001 Stewart, G. W.. Matrix Algorithms, Volume II: Eigensystems. 2001. SIAM.